

Video course
complete DS, ML, DL,
NLP
Thank you sir

sec - 22

Probability Distribution Function for Rvd

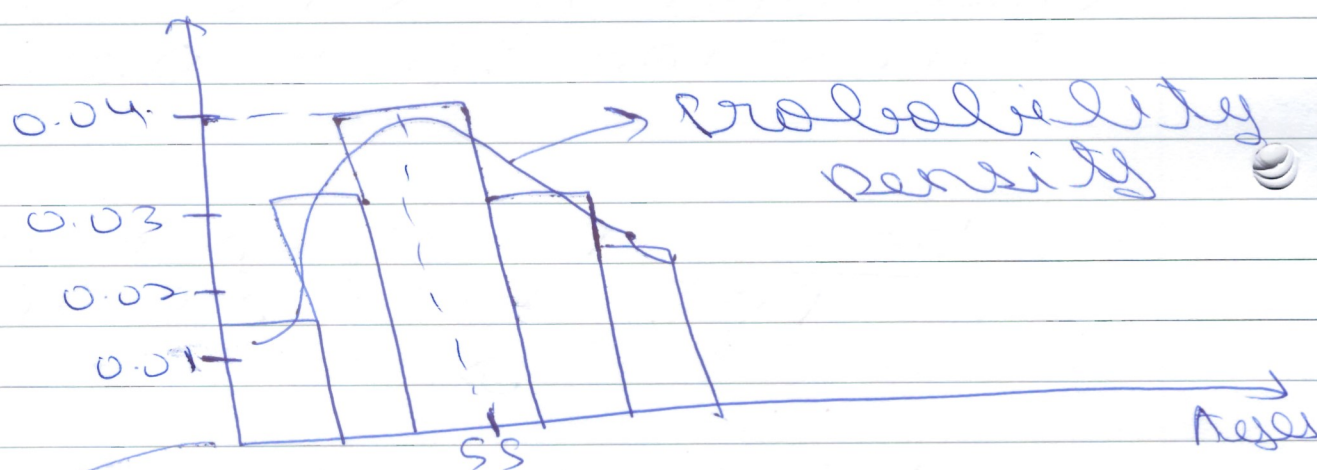
lec-1 Relation b/w
PDF, PMF & CDF
↓

lec-2 types of
distribution

~~The~~ Probability Distribution
Function is the func.
which describes that
how the probabilities
are distributed over
the values of a random
variable.

Ages = { ----- } $\Rightarrow$ Continuous Random Variable:-

Suppose we have an histogram.



Here this curved line becomes the probability density.

Here we are getting probability density equal to 0.04.

Based on the different distributions, we have different types of graphs.

(*)

Now the Probability Distribution Functions are of 2 types:-

(1)

PMF (Probability Mass Function):-

It is used for Discrete Random Variables.

(2)

~~PDF~~ PDF (Probability Density Function)

It is used for continuous Random Variables.

(3) (*)

Now if we discuss the 2 functions carefully, we see that:-

(1)

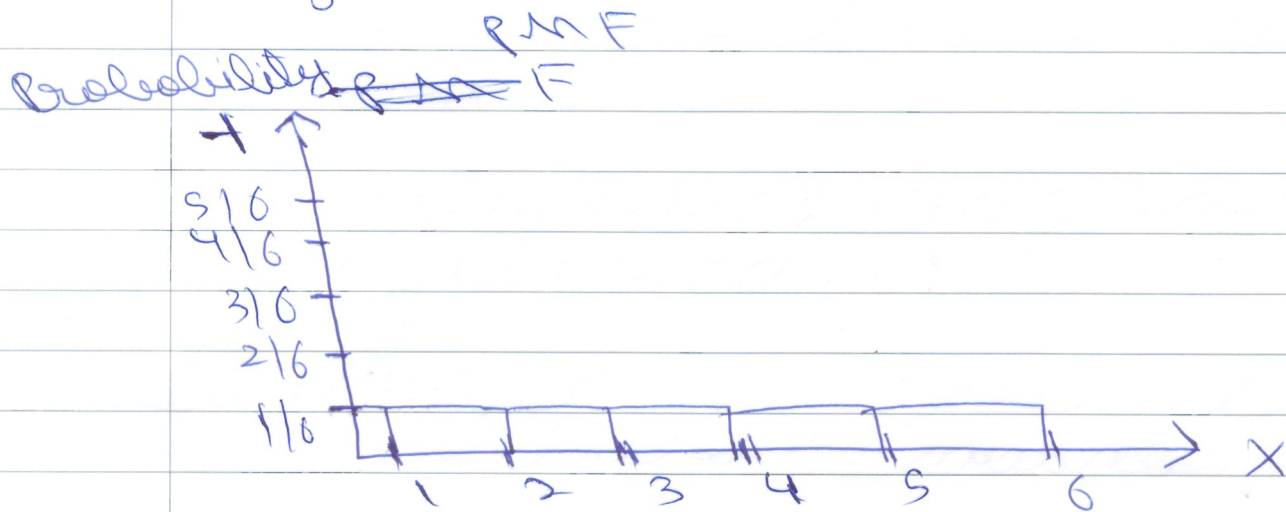
Probability Mass Function
→ discrete

Example :- Rolling a die
we have 6 different outcomes
⇒ {1, 2, 3, 4, 5, 6} P.T.

Here the probability of getting all the numbers is same:-

$$P(1) = P(2) = P(3) = P(4) \\ = P(5) = P(6) = \frac{1}{6}$$

Now Based on the given data, PMF for R:



All probabilities = $P = \frac{1}{6}$

→ Here on y axis we have the graph of Probability

P.T.O.

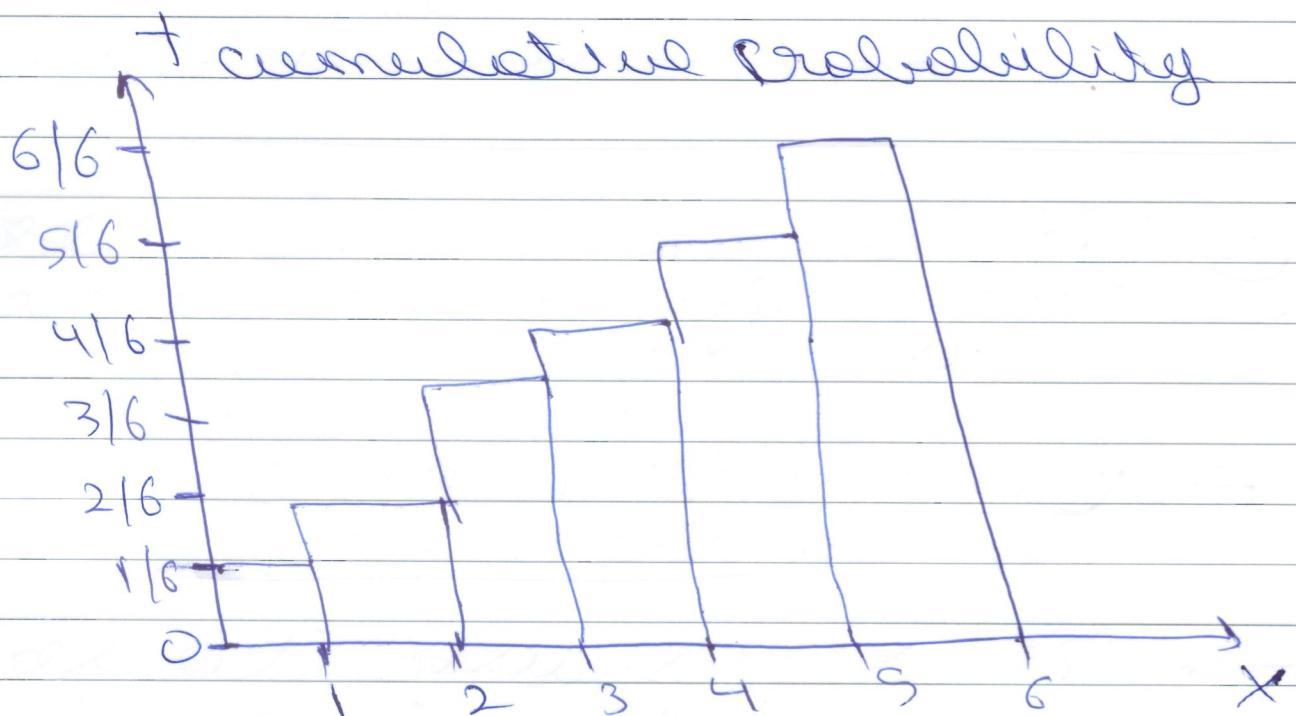
④

Now, we show the function of Cumulative Density Function (CDF)

→ on the y axis, we show Cumulative Probability on the y axis

So basically, overall the probability will keep getting added.

Hence, this is called the Cumulative Density Function (CDF).



Now the probability
for the $x \leq 2$

$$P(x \leq 2) = P(x=1) + P(x=2) \\ = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

The probability of
 $x \leq 6$ can be written

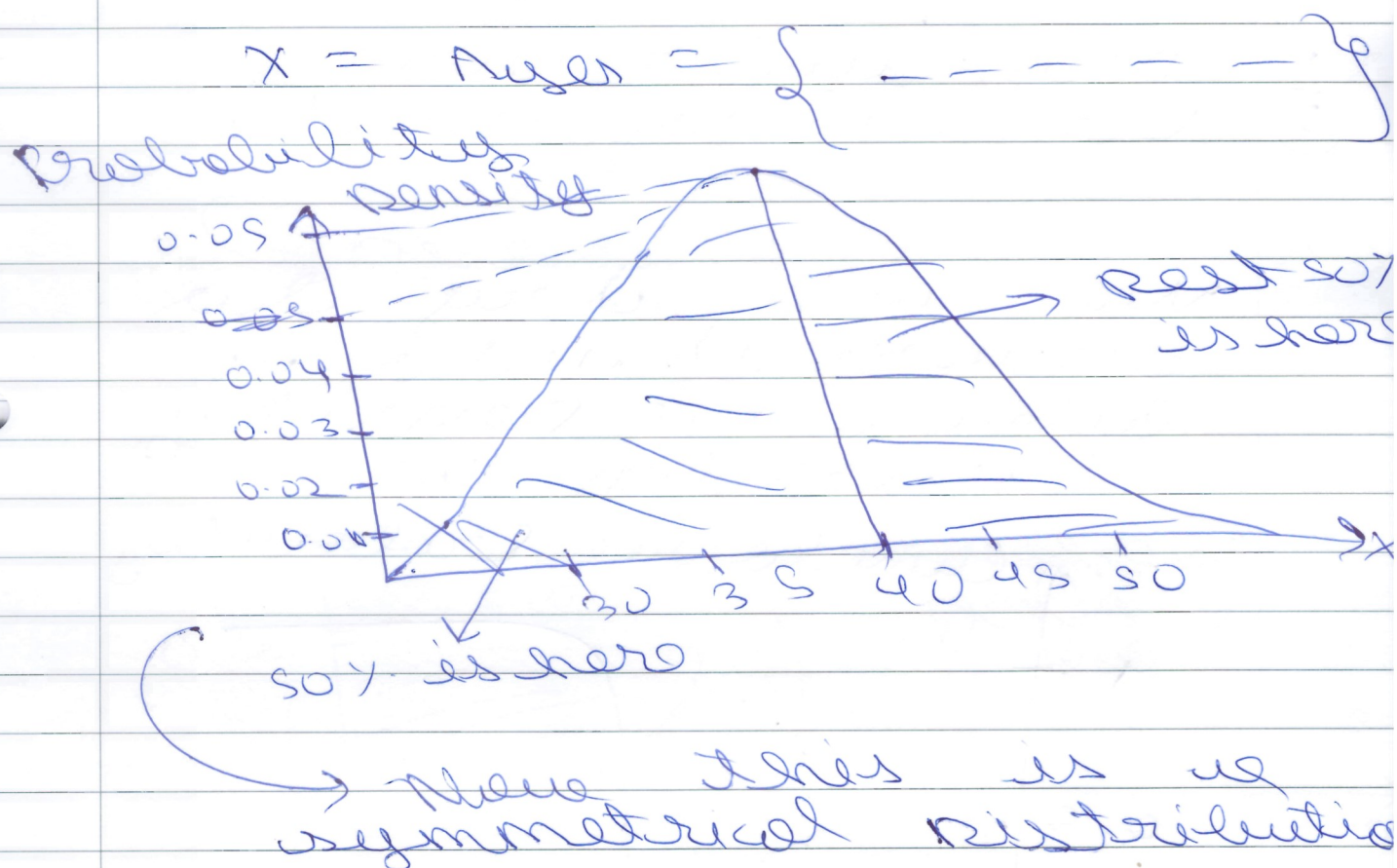
$$P(x \leq 6) = P(x=1) + P(x=2) + \dots + P(x=6) \\ = 1 \text{ Ans}$$

② Now coming to the
second function which

$\Rightarrow$ Probability Density
Function (P.D.F)

This is used to know
the distribution of
continuous random variables.

Suppose we take a graph as:-



Now suppose we want the probability of 40
So here we get as:-

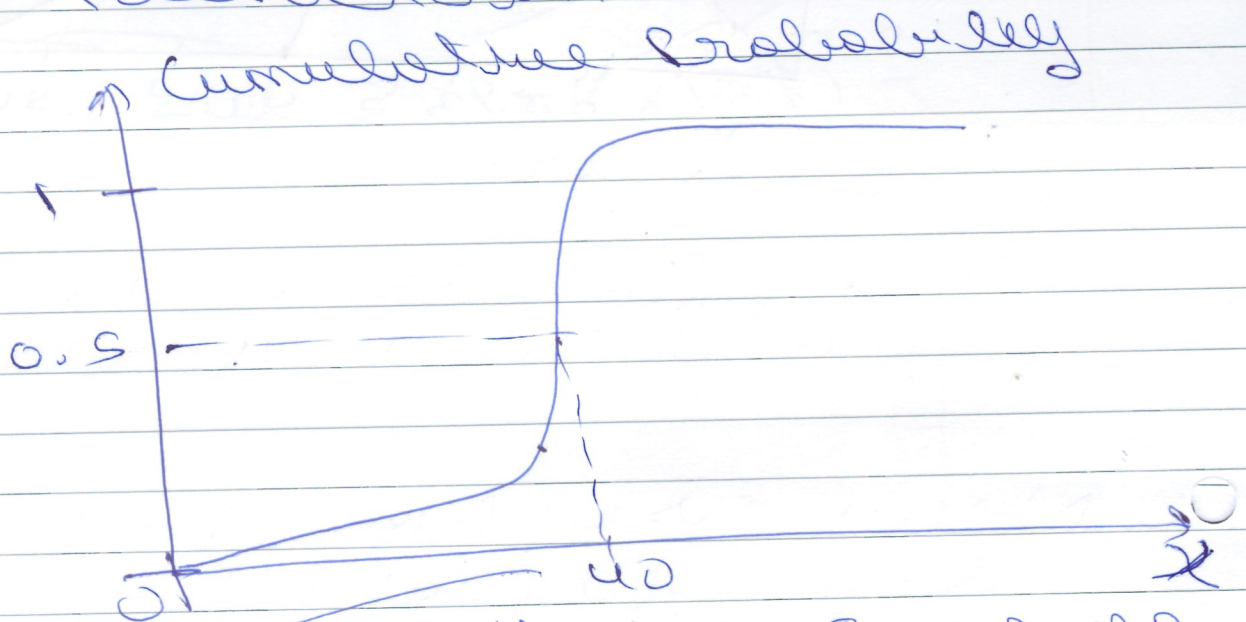
$$P(X=40) = 0.02$$

→ The probability comes out to be 0.02
→ this does not mean this

④ We will be discussing mainly 2 topics:-

- ① Area under Curve
- ② Probability Density.

We will try to convert it into a Cumulative Density Function. We:-



It takes S shape

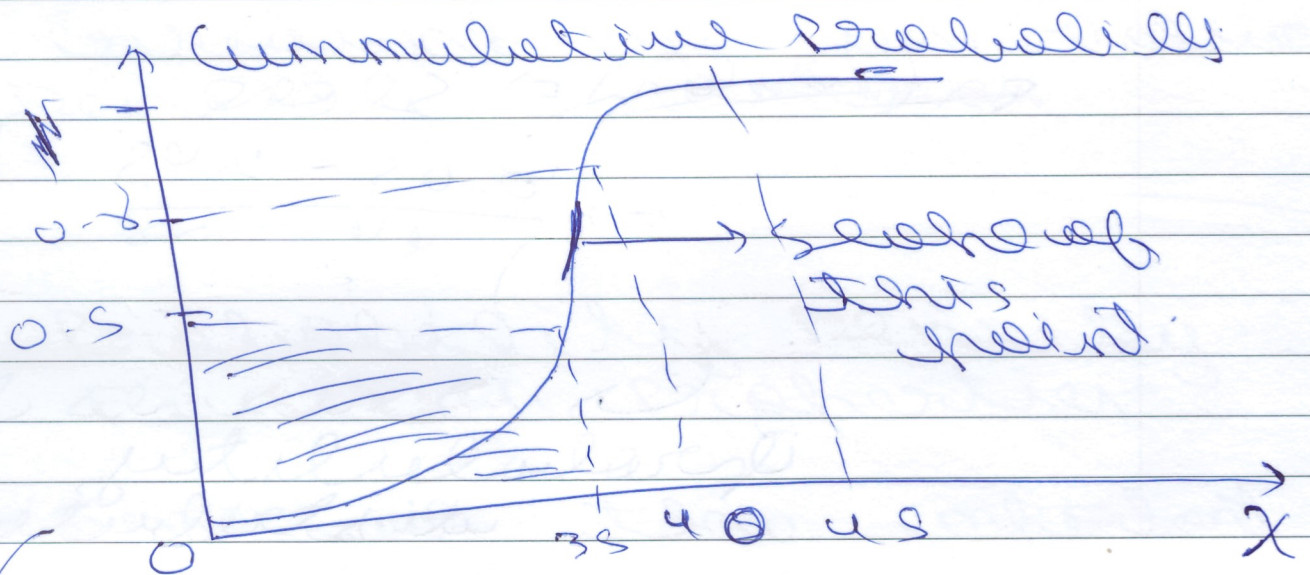
Here the mean is 40

(9)

the probability of
 $X \leq 40 = 0.5 = 50\%$

Now if we plot till
45, we see it is -

→ The graph follows



$$P(X \leq 45) = 0.8 = 80\%$$

→ Now the slope
at this point
is the highest.

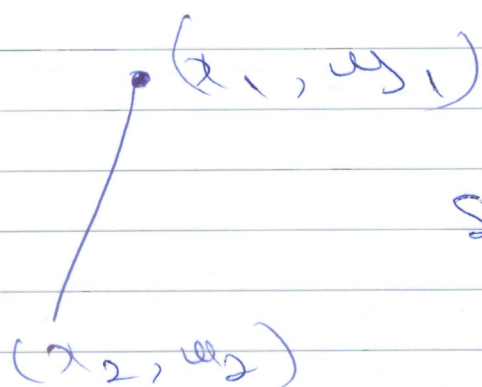
when, suppose we take
for s_0 we see the
slope is not the
minimum.

The points of the
slope are ~~0.05 to 0.1~~
 $0.65 \rightarrow 0.45$

~~$pa(x, y) \Rightarrow$~~ slope

$$\Rightarrow \left\{ \frac{0.65 - 0.45}{44 - 38} \right\}$$

$\rightarrow$ this is the
probability
distribution



$$\text{slope} = \left[\frac{y_2 - y_1}{x_2 - x_1} \right]$$

$\swarrow$
coefficient

$\swarrow$
this is same as
group

so basically the probability density can be mentioned as:-

Probability density:-

It is basically gradient of cumulative density function.

=> Probability Density Function Properties:-

some of the important properties include:-

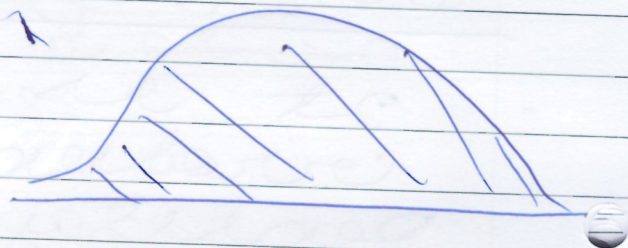
(1) Non Negativity:-
 $f(x) \geq 0$ for all x

(2) The total area under PDF curve is equal to 1.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

So basically, this notation represents area under the PDF curve.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$



This distribution is created by this function $f(x)$

W.r.t the different distribution, $f(x)$ is going to change.

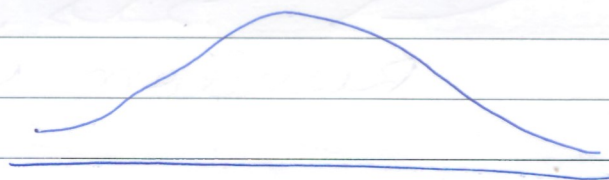
lec - 2Types of Probability Distribution

Suppose we have the different random variables as:-

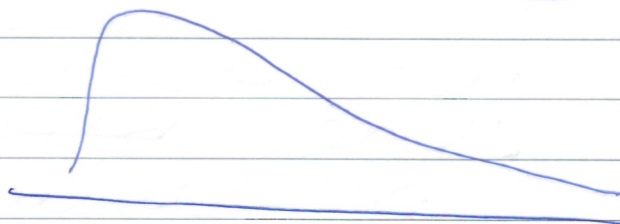
Ages, weight, salary -

the different random variables follow the different distribution as:-

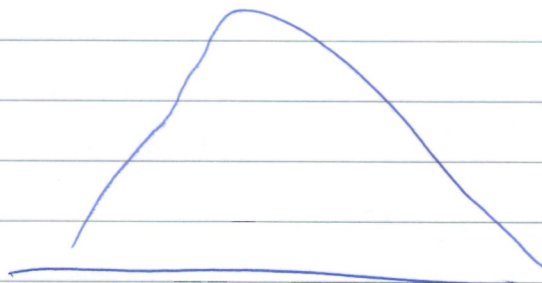
Ages:-



weight:-



Salary:-



Dataset is basically a set of distributions.

(1) Bernoulli Distribution

- Here the outcomes are completely Binary.
- To construct the distribution, we have to use a PMF function.

→ It's a Discrete Random Variable.

(2) We can see the Binary Binomial distribution → PMF

→ Here also PMF is applied.

(3) Normal / Gaussian Distribution $\rightarrow$

$\rightarrow$ 2D created by P.D.F (Probability Distribution Function).

$\rightarrow$ Here there will be assumptions based on which distribution is done.

(4) Poisson Distribution (P.M.F)

(5) Bay Normal Distribution
 $\rightarrow$ Some datasets follow this distribution & uses P.D.F as a function in it.

(6) Uniform Distribution (D.M.F)

Like we have the
data set like:-

→ House Price Prediction
Dataset.

Suppose we have the
features as:-

Size of house → Continuous
Random
Variable
(PDF)

No. of rooms → Discrete
fpmfy

Floor → Discrete

Sea side → Bernoulli
distribution
0, 1
fpmfy

Price → Continuous Random
Variable.
fpmfy

(17)

It helps in making
assumptions for EDA
& feature engineering.